1 PYTHON – Answers

What is the difference between list, tuple, set, dictionary?

- **List**: Ordered, mutable, allows duplicates
- **Tuple**: Ordered, immutable, faster than list
- **Set**: Unordered, mutable, no duplicates
- **Dictionary**: Key–value pairs, mutable, fast lookup

Mutable vs Immutable objects?

- **Mutable**: Can be changed after creation (list, dict, set)
- **Immutable**: Cannot be changed (int, float, string, tuple)

`None` vs `NaN`

- **None**: Python object meaning “no value”
- **NaN**: Numerical missing value (used in NumPy/Pandas)

Shallow copy vs Deep copy

- **Shallow copy**: Copies reference, changes affect original
- **Deep copy**: Copies entire object, fully independent

Lambda function

- Anonymous one-line function
  Example: `lambda x: x*x`

What are decorators?

- Functions that modify other functions without changing their code

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List comprehension

- Shorter way to write loops
  `[x*x for x in range(5)]`

`append()` vs `extend()`

- `append()` → adds one element
- `extend()` → adds multiple elements

`*args` and `**kwargs`

- `*args`: variable positional arguments
- `**kwargs`: variable keyword arguments

How does Python manage memory?

- Uses **automatic garbage collection**
- Reference counting + cyclic garbage collector

2️⃣ SQL – Answers

WHERE vs HAVING

- **WHERE**: filters rows before grouping
- **HAVING**: filters after `GROUP BY`

DELETE vs TRUNCATE vs DROP

- **DELETE**: removes rows, can rollback
- **TRUNCATE**: removes all rows, faster, no rollback
- **DROP**: removes table completely

Primary key vs Foreign key

- **Primary key**: uniquely identifies a record
- **Foreign key**: links to primary key of another table

What is NULL?

- Represents missing or unknown value

Types of joins

- INNER, LEFT, RIGHT, FULL

INNER JOIN vs LEFT JOIN

- **INNER**: matching rows only
- **LEFT**: all left + matching right

Subquery

- Query inside another query

GROUP BY

- Groups rows to apply aggregate functions

Window functions

- Perform calculations without collapsing rows

ROW_NUMBER vs RANK vs DENSE_RANK

- ROW_NUMBER: unique numbers
- RANK: skips numbers
- DENSE_RANK: no skips

2nd highest salary

```
SELECT MAX(salary)

FROM employees

WHERE salary < (SELECT MAX(salary) FROM employees);
```

Indexing

- Improves search speed
  ❌ Slows insert/update operations

3️⃣ STATISTICS – Answers

Mean, Median, Mode

- **Mean**: average
- **Median**: middle value
- **Mode**: most frequent value

Variance & Standard Deviation

- Variance: spread of data
- Std Dev: square root of variance

Correlation vs Covariance

- Correlation: strength + direction
- Covariance: direction only

Normal distribution

- Bell-shaped, symmetric distribution

Central Limit Theorem

- Sample means form a normal distribution

P-value

- Probability of observing result if null hypothesis is true

Hypothesis testing

- Statistical method to test assumptions

Type I & Type II error

- Type I: false positive
- Type II: false negative

Confidence interval

- Range where true parameter lies

Z-test vs T-test

- Z-test: large sample, known variance
- T-test: small sample, unknown variance

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4 MACHINE LEARNING – Answers

What is ML?

- Teaching machines to learn from data

Types of ML

- Supervised, Unsupervised, Reinforcement

Classification vs Regression

- Classification: categories
- Regression: continuous values

Linear Regression assumptions

- Linearity, independence, homoscedasticity, normality

Logistic Regression

- Uses sigmoid to predict probability

Decision Tree splitting

- Based on Gini or Entropy

Random Forest advantage

- Reduces overfitting using multiple trees

KNN – choosing K

- Use elbow method or cross-validation

SVM margin

- Maximum distance between classes

Naive Bayes

- Assumes feature independence

Bias vs Variance

- Bias: underfitting
- Variance: overfitting

Cross-validation

- Splits data into multiple folds

Hyperparameter tuning

- Finding best model parameters

Feature scaling

- Brings features to same scale

Encoding

- Convert categorical to numerical

5 DEEP LEARNING – Answers

ANN

- Network of neurons with weights

Activation function

- Adds non-linearity

Types

- Sigmoid, ReLU, Tanh, Softmax

Why ReLU?

- Avoids vanishing gradient

Backpropagation

- Updates weights using gradients

Vanishing gradient

- Gradients become too small

Dropout

- Randomly disables neurons to prevent overfitting

ANN vs CNN

- CNN handles images better

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6 NLP – Answers

NLP

- Processing human language

Tokenization

- Splitting text into words

Stemming vs Lemmatization

- Stemming: crude chopping
- Lemmatization: dictionary-based

Bag of Words

- Word frequency representation

TF-IDF

- Importance of word in document

Word embedding

- Dense vector representation

Cosine similarity

- Measures angle between vectors

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7 UNSUPERVISED – Answers

Clustering

- Grouping similar data

KMeans steps

- Initialize → assign → update → repeat

Choosing K

- Elbow method

Silhouette score

- Measures cluster quality

Hierarchical clustering

- Tree-based clustering

DBSCAN advantage

- Detects noise, no K needed

PCA

- Dimensionality reduction

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8 METRICS – Answers

Accuracy

- Correct predictions / total

Precision

- Correct positives / predicted positives

Recall

- Correct positives / actual positives

F1-score

- Balance of precision & recall

ROC-AUC

- Model’s class-separation ability

MAE / MSE / RMSE

- Error measurement methods

R^2

- Explained variance

9 EDA – Answers

EDA

- Understanding data patterns

Missing values

- Drop or impute

Outliers

- IQR, Z-score

Feature engineering

- Creating meaningful features

Data leakage

- Using future information

Normalization vs Standardization

- Normalization: 0–1
- Standardization: mean=0, std=1